

14.1: Solving Equations Numerically:

For a continuous function $f(x)$ over interval $[a,b]$, and $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$, this implies there is a root on this interval.

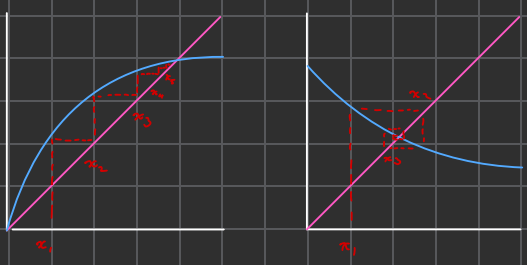
$$f(x) \in C([a,b]) \wedge f(a)f(b) < 0 \Rightarrow \exists x \in (a,b) \mid f(x) = 0$$

This method can then be repeated at smaller intervals to find where the root may be.
This method may break if the function isn't continuous, the root only touches $x=0$, there are several roots close together (can be disproven by finding turning points)

Fixed Point Iteration:

Rearranging Equation $f(x)=0$ to $x = g(x)$ finds the root with line $y=x$
This can then be rearranged to $x_{n+1} = g(x_n)$

For a point x_n, if $g'(x_n) < 0 \Rightarrow$ It will overshoot & form a cobweb diagram
if $g'(x_n) > 0 \Rightarrow$ it will converge uniformly towards the root.



This method only converges to a root, α, iff $|g'(\alpha)| < 1$
This is why seperate rearrangements may be required

Conditions:

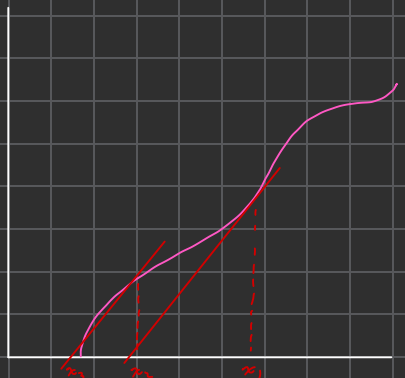
Numerical methods should be used when analytical methods do not function
Accuracy of an answer should be given, and large.
Iterate many, many times
Other rearrangements or methods are sometimes required

14.2: The Newton-Raphson Method:

For a line: $f(x_1) - f(x_2) = f'(x)[x_1 - x_2]$
Given $f(x_2) = 0 \Rightarrow \dfrac{f(x_1)}{f'(x)} = x_1 - x_2$

$$\Rightarrow x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$ This is the Newton-Raphson Method.



Problems with Newton-Raphson

Rate of convergence depends on initial value choice & shape of curve in the neighbourhood of the root. If the initial value is not close the iteration may converge to another root, or diverge.
If done at a turning point, it will not work
Does not work with discontinuities

14.3: Numerical Integration:

For a curve x^2, the area can be approximated using areas of rectangles.





$$A_1 = \delta x \sum_{x=a}^{b-\delta x} f(x) \qquad A_2 = \delta x \sum_{x=a+\delta x}^{b} f(x)$$

Hence:

$$A_1 < \int_a^b x^2\,dx < A_2$$ These are the bounds of the integral.

For higher accuracy, use trapezoids



Hence: $A_T = \frac{1}{2}\delta x\left(f(a)+f(b)+2\sum_{a+\delta x}^{b-\delta x} f(x)\right)$

If $\frac{d^2y}{dx^2} > 0 \Rightarrow \int_a^b f(x)\,dx < A_T$

If $\frac{d^2y}{dx^2} < 0 \Rightarrow \int_a^b f(x)\,dx > A_T$

If $\frac{d^2y}{dx^2}$ changes, you cannot be sure.